

UNNS Color Confinement - Regime Synthesis Note

Admissibility-by-closure, repair thresholds, and localized route geometry

Status: synthesis of three formal diagnostics, not a derivation of QCD confinement.

1. Purpose

This report combines three formal diagnostics: First Confinement Diagnostic, Repair-Threshold Analysis, and Flux-Tube Separation Extension. The aim is to state what color confinement now adds to the UNNS regime-synthesis program while preserving the boundary between structural interpretation and full QCD derivation.

2. Core synthesis

Colored constituents may be internally real, but isolated colored externalization is not the admissible output. The observable route is tensioned, localized, and repaired into color-neutral composite channels.

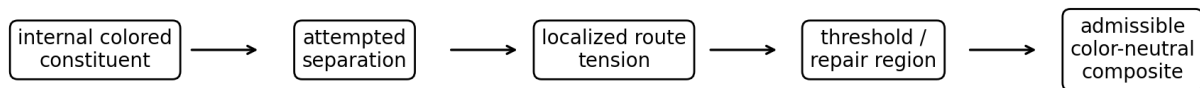
internal colored coordinate -> attempted route extension -> localized route tension -> repair-threshold window -> admissible color-neutral composite route

3. Diagnostics synthesized

Diagnostic	Main data object	UNNS object	Boundary
01 First Confinement	static-source spectrum + single flux-tube profile	route extension, tension proxy, localized geometry	V0,V1,V2 reconstructed; flux profile pointwise
02 Repair Threshold	r_c/r_cs windows + gap behavior	repair markers and channel competition	thresholds reported; gaps reconstructed
03 Flux-Tube Extension	Baker profiles across separations	route localization tracked across d	main trend uses QC-accepted FULL/NP rows

4. Diagnostic 01 - first confinement map

Static-source rows: 57; static separation range: 0.7069-1.6065 fm. Static data status: model reconstruction from reported parameters, not raw measured point data. The first flux-tube pointwise profile has 31 rows at source separation 0.738309 fm; peak Ex FULL = 0.343135 GeV² and peak Ex NP = 0.244566 GeV².



UNNS working map: route extension -> boundary pressure -> repair into closed admissible state

Figure 1. Working UNNS sequence extracted from the first diagnostic.

5. Diagnostic 02 - repair-threshold window

Marker	Channel	r [fm]	nearest r [fm]	gap V1-V0 [GeV]	gap V2-V1 [GeV]
r_c	light_static_light_threshold	1.224	1.22094	0.08970	0.04594
r_cs	strange_static_strange_threshold	1.293	1.28520	0.08642	0.04639

The repair-threshold diagnostic extracts a route-repair object: stretched color route -> threshold-window approach -> competing screened channel -> repaired admissible color-neutral composite route.

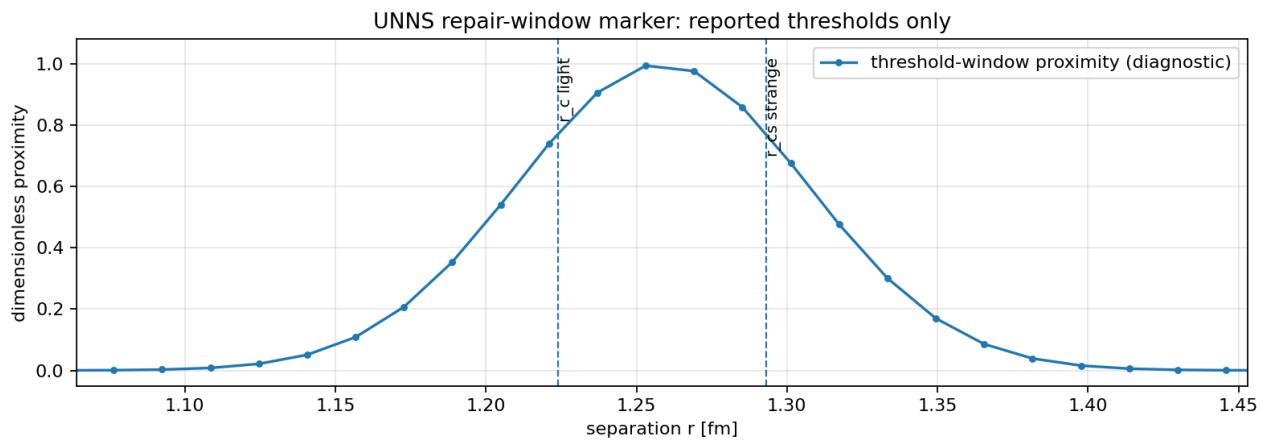


Figure 2. Repair-window marker using reported thresholds.

6. Diagnostic 03 - flux-tube separation extension

The Baker extension loaded 1090 pointwise rows across 41 profiles and 10 source files, with no missing files. QC retained 30 FULL/NP rows for trend analysis, placed 9 rows under separate review, and excluded 2 single-point summary rows. The trusted trend range is 0.723-1.060 fm.

Component	rows	d range [fm]	peak range [GeV^2]	FWHM range [fm]	mean area
FULL	15	0.723-1.060	0.195-0.387	0.530-0.663	0.195
NP	15	0.723-1.060	0.126-0.274	0.544-0.723	0.152

Descriptive slopes inside trusted rows only (not a fitted law):

Component	Quantity	slope vs d	correlation
FULL	peak	-0.4914	-0.964
FULL	FWHM	0.3764	0.963
FULL	area	-0.1898	-0.969
NP	peak	-0.2739	-0.888
NP	FWHM	0.3963	0.900
NP	area	-0.0813	-0.654

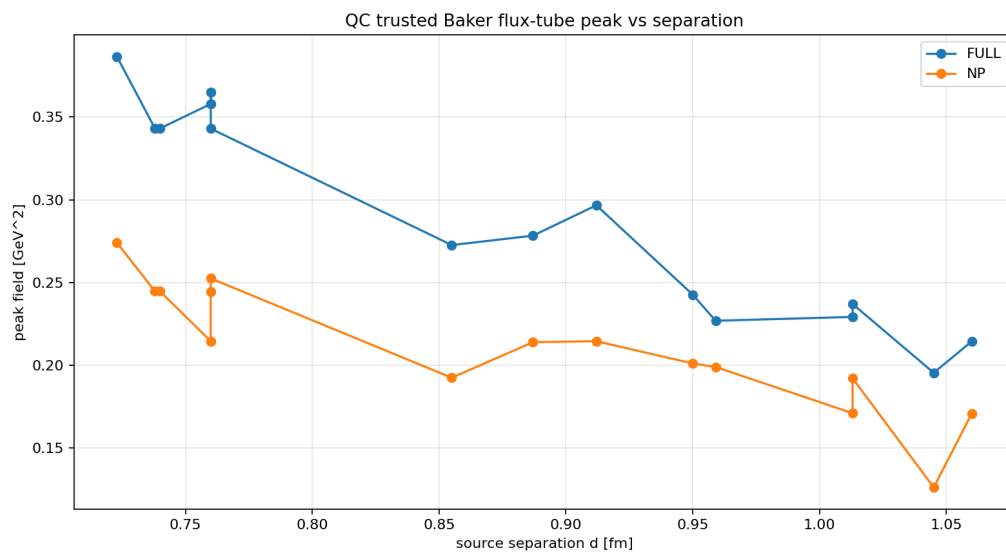


Figure 3. Trusted Baker peak trend (FULL/NP only).

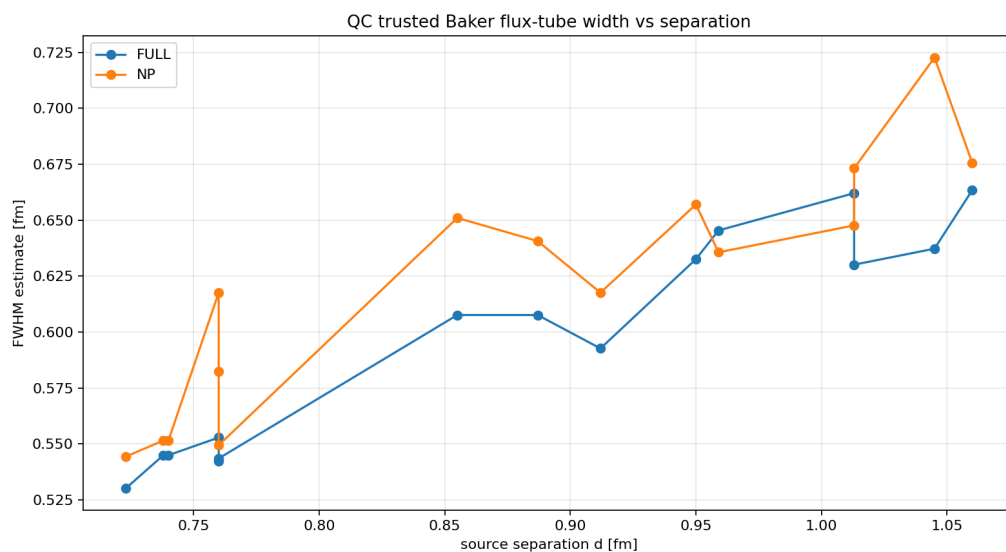


Figure 4. Trusted Baker width trend (FULL/NP only).

7. Review-separately boundary

The nine non-trend rows are not discarded, but they are not mixed into the main trend. They are kept as qualitative string-breaking, large-distance control, normalization-later, or outlier-note rows. This prevents accidental claims of route broadening from mixed source conventions.

Review decision	count
KEEP_SEPARATE_QUALITATIVE_STRING_BREAKING	3
EXCLUDE_FROM_INTERPRETATION_FOR_NOW	2
NORMALIZE_LATER_QUALITATIVE_ONLY	2
KEEP_SEPARATE_LARGE_DISTANCE_CONTROL	1
EXCLUDE_FROM_TREND_KEEP_AS_OUTLIER_NOTE	1

8. What color confinement now adds to UNNS

- Color confinement is a grounded benchmark for internal reality vs external admissibility.
- Static-source separation provides a route-extension coordinate.
- Static spectrum and gaps provide route-tension and channel-competition proxies.
- String-breaking thresholds provide repair-window markers.
- Flux-tube profiles provide localized route geometry across separation.
- The work aligns with Charge Boundary-Route Preservation and threshold-repair logic without claiming a derivation of QCD confinement.

9. Supported now / not yet supported

Supported now	Not yet supported
admissibility-by-closure benchmark	derivation of QCD confinement from UNNS
route extension and localized geometry proxies	final quantitative UNNS confinement law
repair threshold markers r_c and r_{cs}	raw measured V_0, V_1, V_2 GEVP tables
trusted FULL/NP flux-tube trend 0.723-1.060 fm	large-distance broadening law from PAPER_DEFINED rows

10. Next work item

The next work item is not another broad data import. It is a threshold-local boundary-pressure proxy. A provisional symbolic form may be drafted only as: $B_{color}(r) = \text{normalized route-tension spectrum} + \text{gap-compression term} + \text{localized-profile width/peak term}$. It should not be fitted until measured or digitized static levels are available near r_c and r_{cs} .

11. Final synthesis statement

A colored constituent can be internally real while remaining externally inadmissible as an isolated object. The admissible observable is produced by route closure. Attempted separation produces localized tension and threshold-channel competition; the continuation is repair into color-neutral composite routes, not free color externalization.